

Lesson 2B: Solving Equations with Radical Expression

Criteria	Exceeds Standard (4)	Meets Standard (3)	Approaching Standard (2)	Below Standard (1)
Isolating the Radical	Correctly isolates the radical on one side of the equation immediately.	Isolates the radical with minor errors.	Attempts to isolate, but makes significant errors.	Cannot identify how to isolate the radical.
Solving by Powers	Correctly raises both sides to the nth power (e.g., squaring) to eliminate all radicals.	Squares both sides, but makes minor algebra errors in the process.	Attempts to square but does not square both sides correctly or entirely.	Cannot determine the necessary power to remove the radical.
Algebraic Solving	Correctly solves the resulting polynomial/linear equation for x.	Solves for x but with minor computational errors.	Makes significant errors in solving the resulting equation.	Cannot solve the resulting equation.
Extraneous Solutions	Accurately identifies all solutions and removes all extraneous solutions.	Identifies most solutions, misses one extraneous solution.	Does not check for extraneous solutions.	Fails to recognize that solutions may be extraneous.
Verification (Checking)	Extensively verifies answers by plugging them back into the original equation.	Partially verifies solutions.	Checks solutions but makes errors in substitution.	No verification step performed.